# CONTENT

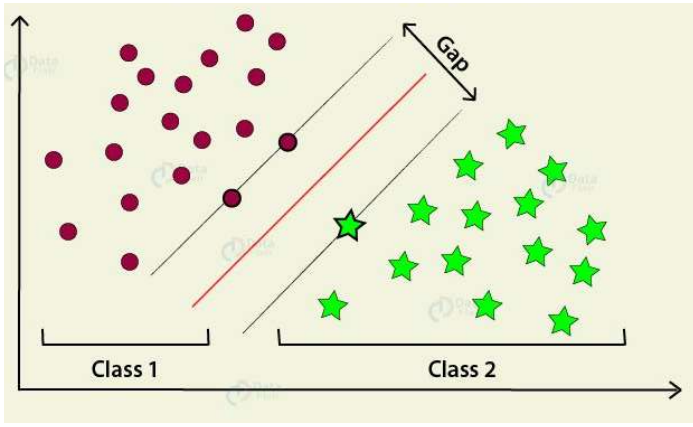
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Think Ahead, Beyond Boundaries


1. Support Vector Machine Algorithm
2. How Does Support Vector Machine Work?
3. What Is Support Vector Regression?
4. What Is Soft-margin Support Vector Machine?
5. Advantages and Disadvantages of Support Vector Machines
6. Applications of Support Vector Machine

# SUPPORT VECTOR MACHINE ALGORITHM

- **Support Vector Machines** are a type of supervised machine learning algorithm that provides data analysis for classification and regression analysis. While they can be used for regression, ***SVM is mostly used for classification***.
- We carry out plotting in the n-dimensional space. The value of each feature is also the value of the specific coordinate. Then, we find the ideal ***hyperplane that differentiates between the two classes***.

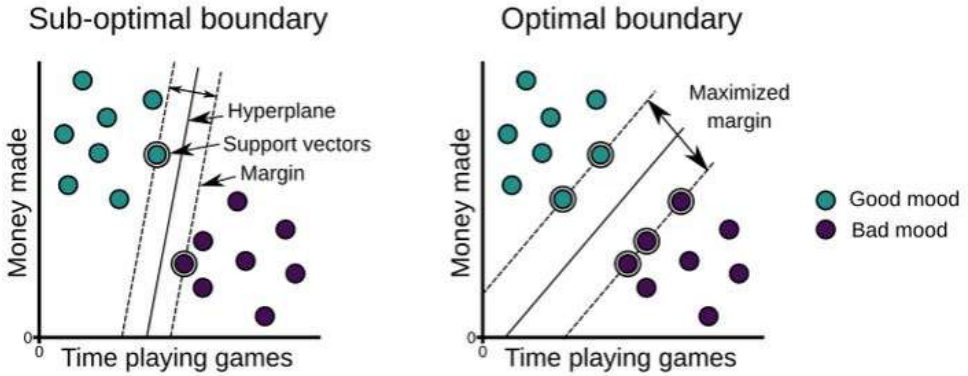
The aim of an SVM algorithm is to maximize this very margin. When the margin reaches its ***maximum***, the ***hyperplane*** becomes the optimal one.

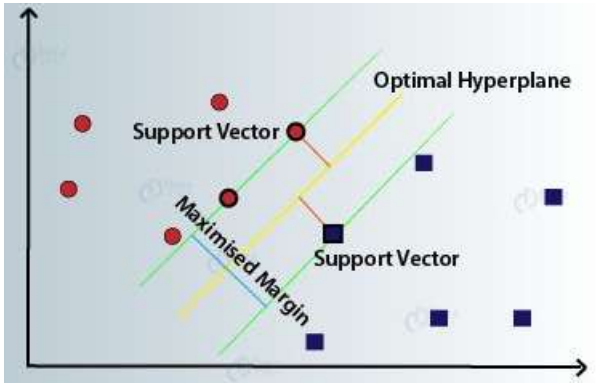



Source: Jain, R. (2020). *Support Vector Machine (SVM)* [online], viewed May 24 2022, from: <https://medium.com/@rishabhjain9440/support-vector-machine-svm-f23cece26419>

# SUPPORT VECTOR MACHINE ALGORITHM

- According to SVM, we have to find the points that lie closest to both the classes. These points are known as **support vectors.**
- In the next step, we find the proximity between our dividing plane and the support vectors. The distance between the points and the dividing line is known as **margin**.
- The aim of an SVM algorithm is to maximize this very margin. When the margin reaches its ***maximum***, the ***hyperplane*** becomes the optimal one.



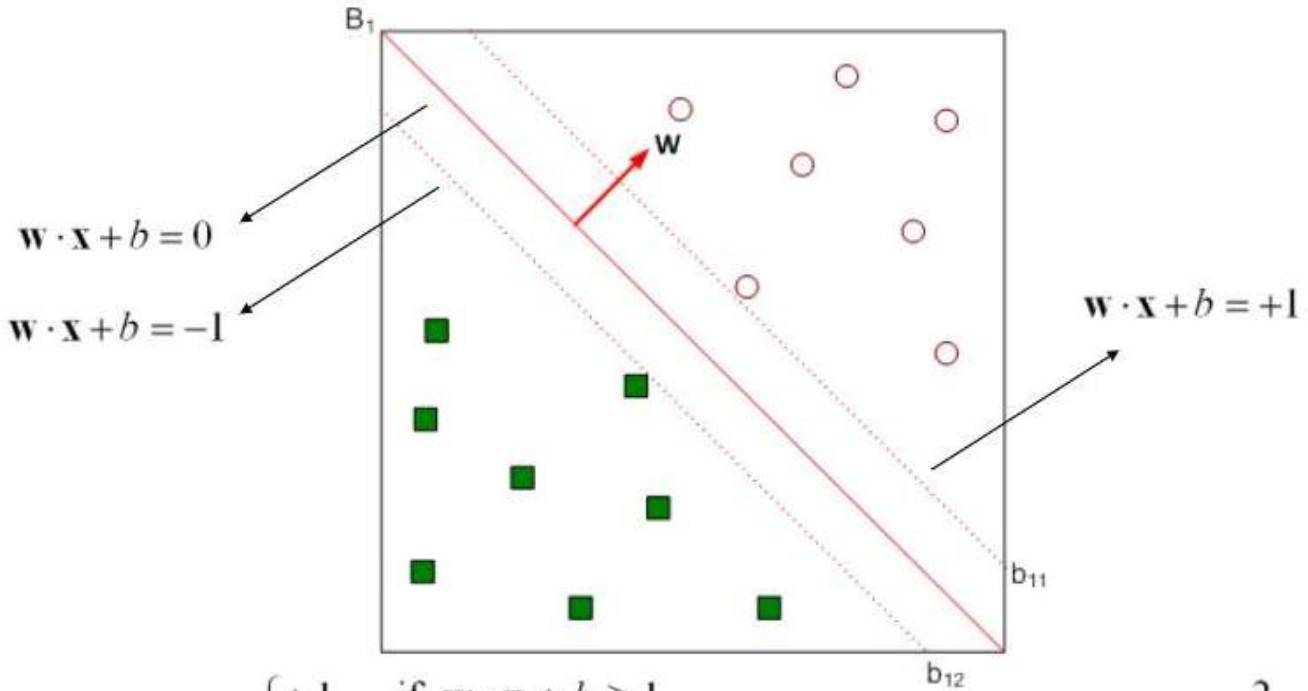




Source:
Top image: Rhys, H. I. (2020). *Machine Learning with R, the tidyverse, and mlr.* USA: Manning Publications
Bottom image: Shashi Rekha G et al. (2020). *Analysis of Fine-Grained Air Quality and Prediction of Air Pollution* [online], viewed May 24 2022, from: <https://www.irjet.net/archives/V7/i5/IRJET-V7I51165.pdf>

# HOW DOES SVM WORK?

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Depict:




$$y_i = f(\mathbf{x}) = \begin{cases} +1 & \text{if } \mathbf{w}\cdot\mathbf{x}+b \geq 1 \\ -1 & \text{if } \mathbf{w}\cdot\mathbf{x}+b \leq -1 \end{cases}$$

$$\text{margin} = \frac{2}{\|\mathbf{w}\|}$$

# HOW DOES SVM WORK?

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Depict:

**Lagrangian function** of a support vector machine

$$L(\mathbf{w}, b, \alpha_i) = \tfrac{1}{2}\mathbf{w}'\mathbf{w} - \sum_{i=1}^{n} \alpha_i \left[y_i(\mathbf{w}'\mathbf{x}_i - b) - 1\right]$$

support vector machine **primal problem**

$$\begin{aligned} \min \quad & \tfrac{1}{2}\|\mathbf{w}\|_2^2 \\ \text{s.t.} \quad & y_i(\mathbf{w}'\mathbf{x}_i - b) \geq 1 \quad \forall i = 1 \ldots n \end{aligned}$$

$$\nabla_{\mathbf{w}} L(\mathbf{w}, b, \alpha_i) = 0: \quad \mathbf{w} = \sum_{i=1}^{n} \alpha_i y_i \mathbf{x}_i$$

$$\nabla_{b} L(\mathbf{w}, b, \alpha_i) = 0: \quad \sum_{i=1}^{n} \alpha_i y_i = 0$$

support vector machine **dual problem**

$$\begin{aligned} \max \quad & -\frac{1}{2} \sum_{i=1}^{n} \sum_{j=1}^{n} \alpha_i \alpha_j y_i y_j \mathbf{x}_i' \mathbf{x}_j + \sum_{i=1}^{n} \alpha_i \\ \text{s.t.} \quad & \sum_{i=1}^{n} \alpha_i y_i = 0, \\ & \alpha_i \geq 0, \quad \forall i = 1 \ldots n \end{aligned}$$

# HOW DOES SVM WORK?

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**SVM Steps in Details**

**Step 1:** SVM algorithm predicts the classes. One of the classes is identified as 1 while the other is identified as -1.

**Step 2:** As all machine learning algorithms convert the business problem into a mathematical equation involving unknowns. These unknowns are then found by converting the problem into an optimization problem. As optimization problems always aim at maximizing or minimizing something while looking and tweaking for the unknowns, in the case of the SVM classifier, a loss function known as the hinge loss function is used and tweaked to find the maximum margin.

**Step 3:** For ease of understanding, this loss function can also be called a cost function whose cost is 0 when no class is incorrectly predicted. However, if this is not the case, then error/loss is calculated. The problem with the current scenario is that there is a trade-off between maximizing margin and the loss generated if the margin is maximized to a very large extent. To bring these concepts in theory, a regularization parameter is added.

# HOW DOES SVM WORK?

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**SVM Steps in Details**

**Step 4:** As is the case with most optimization problems, weights are optimized by calculating the gradients using advanced mathematical concepts of calculus viz. partial derivatives.

**Step 5:** The gradients are updated only by using the regularization parameter when there is no error in the classification while the loss function is also used when misclassification happens.

**Step 6:** When the loss becomes a minimum value, the learning will be terminated.

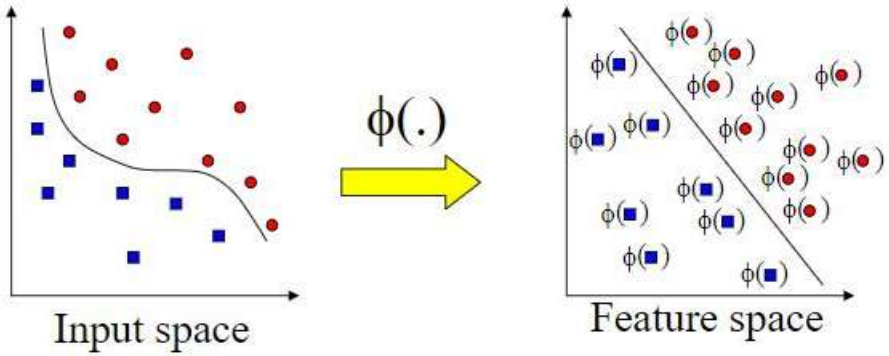
# HOW DOES SVM WORK?

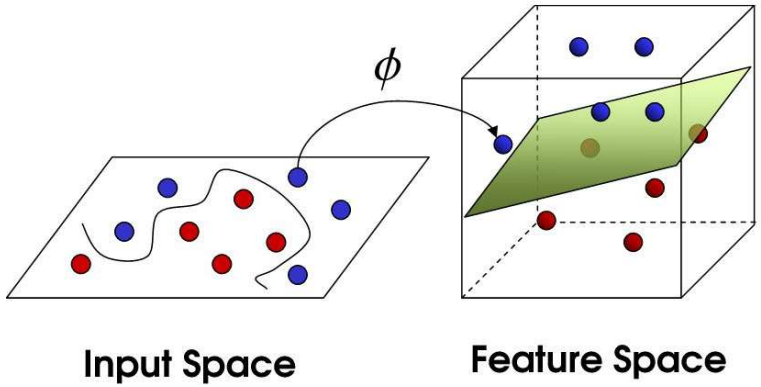
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Depict:

$$k(\mathbf{x}, \mathbf{z}) = \langle \phi(\mathbf{x}), \phi(\mathbf{z}) \rangle$$

$$\phi(\mathbf{x}) = \phi\left(\begin{pmatrix} x_1 \\ x_2 \end{pmatrix}\right) = \begin{pmatrix} x_1^2 \\ \sqrt{2}\, x_1 x_2 \\ x_2^2 \end{pmatrix}$$

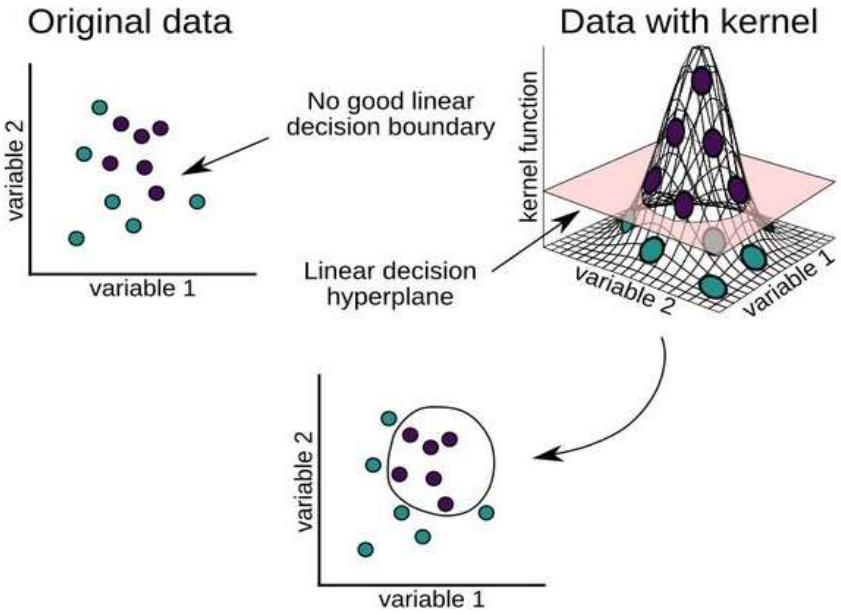


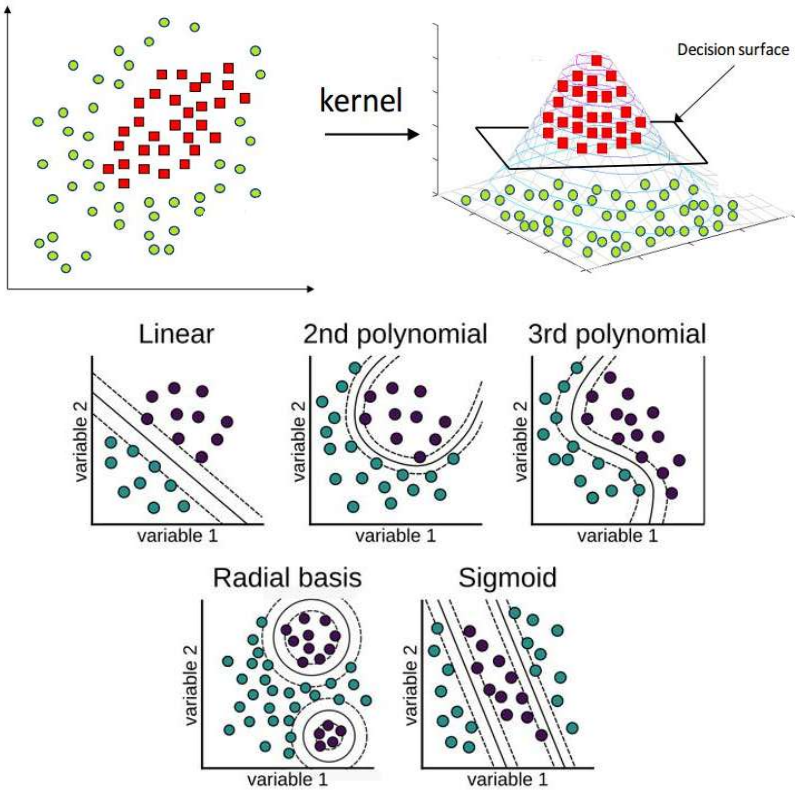




Source:
Top image: Bishop, C. (2006). *Pattern Recognition and Machine Learning*. USA: Springer New York
Bottom image: Saxena, S. (2021). *Beginner's Guide to Support Vector Machine(SVM)* [online], viewed May 24 2022, from: <https://www.analyticsvidhya.com/blog/2021/03/beginners-guide-to-support-vector-machine-svm/>

# HOW DOES SVM WORK?

Depict:




Source (from left to right, from top to bottom):
Image 1: Rhys, H. I. (2020). *Machine Learning with R, the tidyverse, and mlr*. USA: Manning Publications
Image 2: Sharma, S. (2019). *Kernel Trick in SVM* [online], viewed May 24 2022, from: <https://medium.com/analytics-vidhya/how-to-classify-non-linear-data-to-linear-data-bb2df1a6b781>
Image 3: Rhys, H. I. (2020). *Machine Learning with R, the tidyverse, and mlr*. USA: Manning Publications

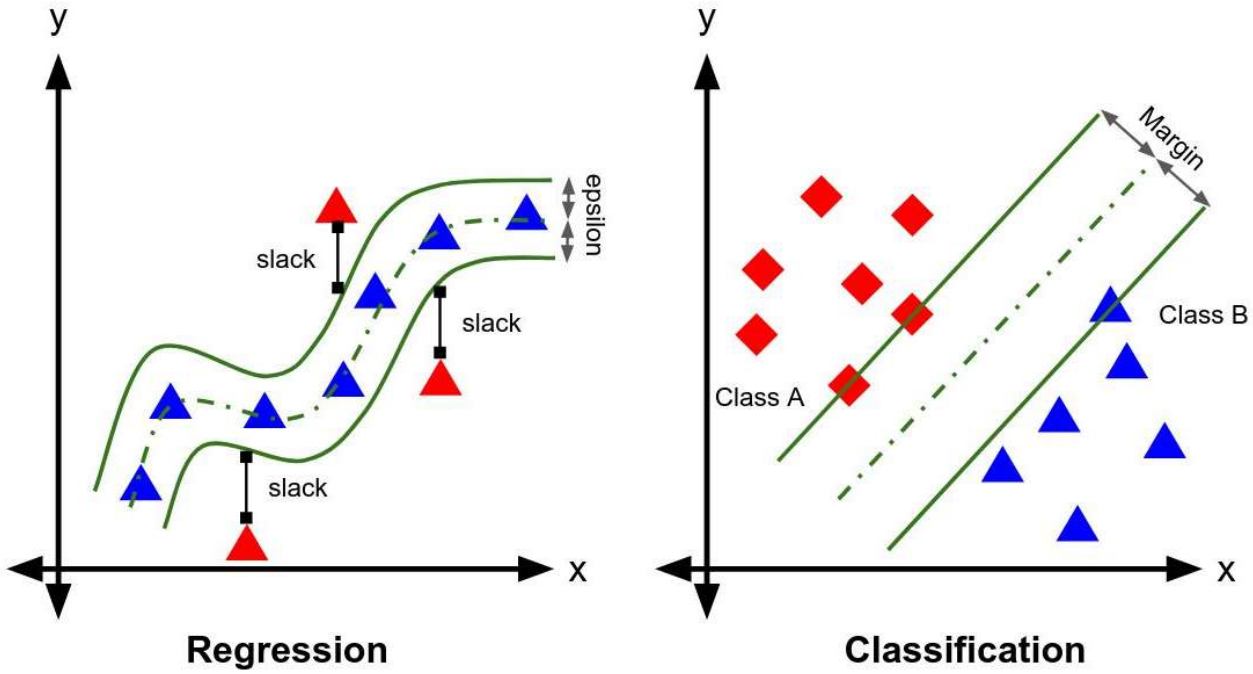
# WHAT IS SUPPORT VECTOR REGRESSION?

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- **Support Vector Regression** is a supervised learning algorithm that is used to predict discrete values. Support Vector Regression uses the same principle as the SVMs. The basic idea behind SVR is to find the best fit line. In SVR, the best fit line is the hyperplane that has the maximum number of points.
- Unlike other Regression models that try to minimize the error between the real value and predicted value, the SVR tries to fit the best line within a threshold value. The threshold value is the distance between the hyperplane and boundary line. The fit time complexity of SVR is more than quadratic with the number of samples, which makes it hard to scale to datasets with more than a couple of 10000 samples.

# WHAT IS SUPPORT VECTOR REGRESSION?

- SVR




Source: Achsan, B. M. (2019). *Support Vector Machine: Regression* [online], viewed May 24 2022, from: <https://medium.com/it-paragon/support-vector-machine-regression-cf65348b6345>

# WHAT IS SUPPORT VECTOR REGRESSION?

- Find a function, f(x), with at most Ɛ-deviation from the target y

The problem can be written as a convex optimization problem

$$\min \frac{1}{2} \| \mathbf{w} \|^2$$

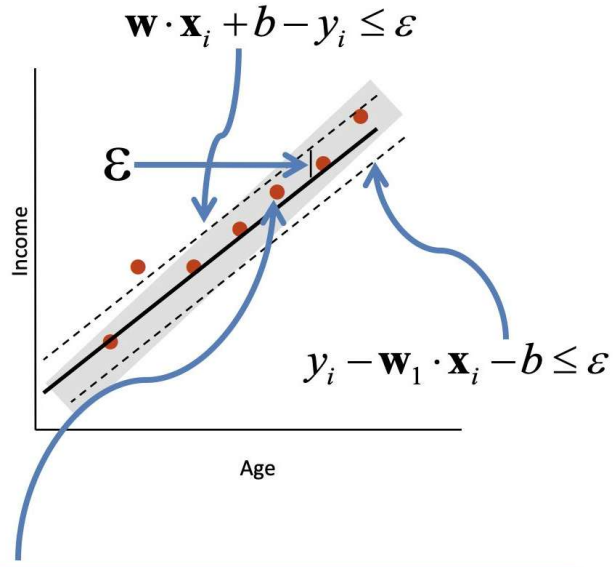
$$s.t.\ y_i - \mathbf{w}_1 \cdot \mathbf{x}_i - b \le \varepsilon;$$

$$\mathbf{w}_1 \cdot \mathbf{x}_i + b - y_i \le \varepsilon;$$

C: trade off the complexity

What if the problem is not feasible?

We can introduce slack variables (similar to soft margin loss function).




We do not care about errors as long as they are less than ε

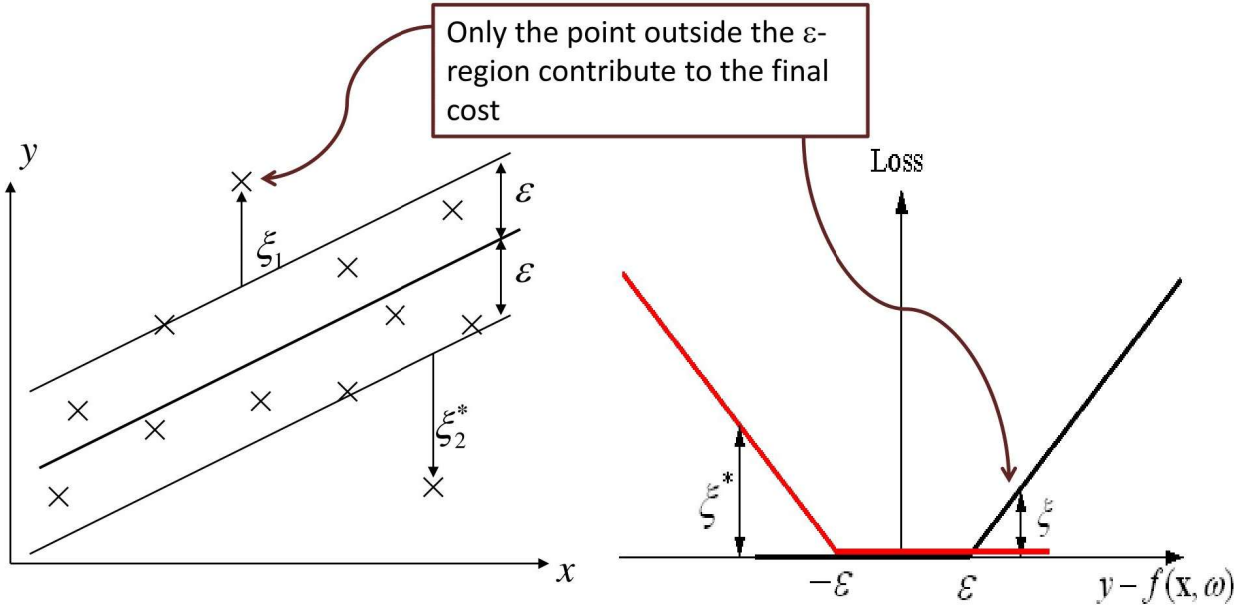
Source: Singh, R. (2019). *Mathematics behind Support Vector Machine* [online], viewed May 24 2022, from: <https://medium.com/analytics-vidhya/mathematics-behind-support-vector-machine-4e8130b83840>

# WHAT IS SUPPORT VECTOR REGRESSION?

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- SVR

Assume linear parameterization $f(\mathbf{x}, \omega) = \mathbf{w} \cdot \mathbf{x} + b$




$$L_\varepsilon(y, f(\mathbf{x}, \omega)) = \max\left(|\, y - f(\mathbf{x}, \omega)\,| - \varepsilon, 0\right)$$

9

Source: Stack Exchange Users. *How should we interpret this figure that relates the perceptron criterion and the hinge loss?* [online], viewed May 24 2022, from: <https://ai.stackexchange.com/questions/24767/how-should-we-interpret-this-figure-that-relates-the-perceptron-criterion-and-th>

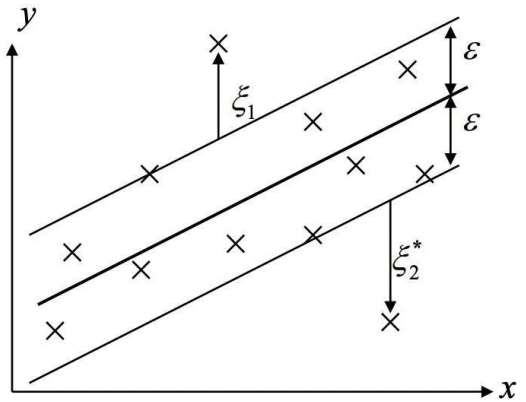
# WHAT IS SUPPORT VECTOR REGRESSION?

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- SVR Soft Margin

Given training data

$$(\mathbf{x}_i, y_i) \quad i = 1, \ldots, m$$




Minimize

$$\frac{1}{2} \| \mathbf{w} \|^2 + C \sum_{i=1}^{m} (\xi_i + \xi_i^*)$$

Under constraints

$$\begin{cases} y_i - (\mathbf{w} \cdot \mathbf{x}_i) - b \leq \varepsilon + \xi_i \\ (\mathbf{w} \cdot \mathbf{x}_i) + b - y_i \leq \varepsilon + \xi_i^* \\ \xi_i, \xi_i^* \geq 0, i = 1, \ldots, m \end{cases}$$

Source: Stack Exchange Users. *Support vector machines and regression* [online], viewed May 24 2022, from: <https://stats.stackexchange.com/questions/13194/support-vector-machines-and-regression>

# WHAT IS SOFT-MARGIN SVM?

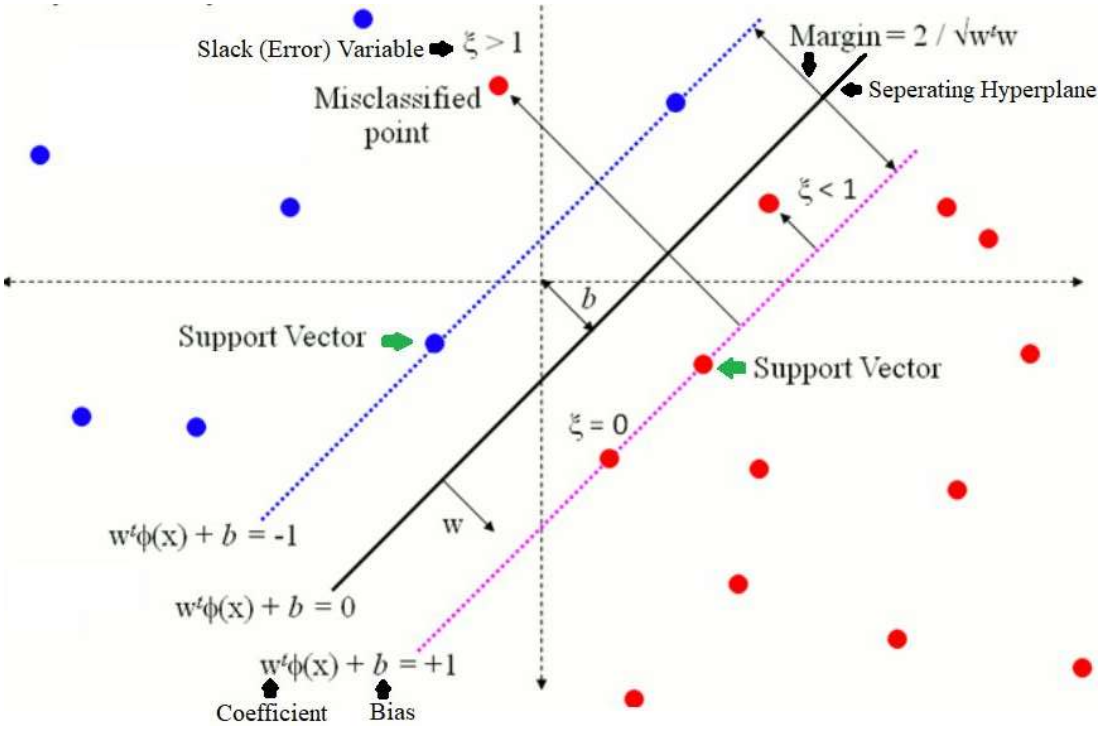
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- If the dataset is noisy (with some overlap in positive and negative samples), there will be some error in classifying them with the hyperplane.
- In the latter case, the objective will be to minimize the errors in classification along with maximizing the margin, and the problem becomes a **soft-margin SVM.**
- A slack variable per training point is introduced to include the classification errors (for the miss-classified points in the training dataset) in the objective. This can also be thought of as adding regularization.
- The optimization problem is quadratic in nature, since it has a quadratic objective with linear constraints.

# WHAT IS SOFT-MARGIN SVM?

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- Depict




Source: Dey, S. (2018). *Implementing a Soft-Margin Kernelized Support Vector Machine Binary Classifier with Quadratic Programming in R and Python* [online], viewed May 24 2022, from: <https://sandipanweb.wordpress.com/2018/04/23/implementing-a-soft-margin-kernelized-support-vector-machine-binary-classifier-with-quadratic-programming-in-r-and-python/>

# WHAT IS SOFT-MARGIN SVM?

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- Depict

**The Primal Problem**

$$min \quad \frac{1}{2}w^T w + C\sum_{i=1}^{m}\xi_i$$

$$s.t. \quad y_i(wx_i + b) \geq 1 - \xi_i$$

$$\xi_i \geq 0$$

*The primal constraints*

*The dual variables (lagrange multipliers)*

**From Primal to Dual**

The Lagrangian

$$\mathcal{L}(w, \xi, b, \alpha) = \frac{1}{2}w^T w + C\sum_{i=1}^{m}\xi_i + \sum_{i=1}^{m}\alpha_i\left[1 - \xi_i - y_i(wx_i + b)\right] + \sum_{i=1}^{m}\mu_i(-\xi_i)$$

*(minimize w.r.t. w, b, ξ for a fixed α)*

*The KKT conditions*

$$\frac{\partial\mathcal{L}}{\partial w} = w - \sum_{i=1}^{m}\alpha_i y_i x_i = 0 \Rightarrow w = \sum_{i=1}^{m}\alpha_i y_i x_i$$

$$\frac{\partial\mathcal{L}}{\partial b} = \sum_{i=1}^{m}\alpha_i y_i = 0$$

$$\frac{\partial\mathcal{L}}{\partial \xi_i} = C - \alpha_i - \mu_i = 0 \Rightarrow \mu_i = C - \alpha_i$$

$$\mu_i \geq 0 \Rightarrow \alpha_i \leq C$$

**The Dual Problem**

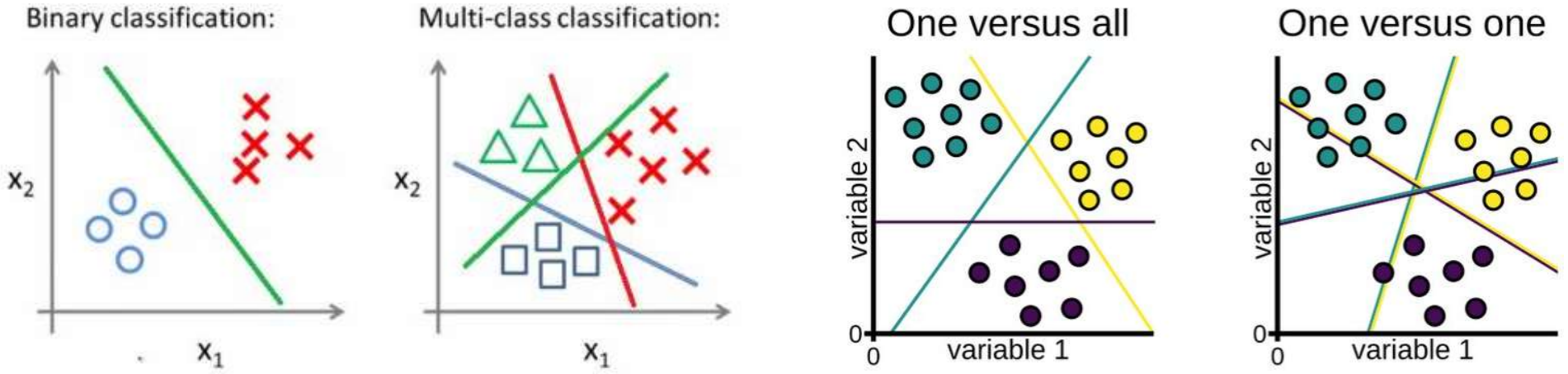
$$\max_{\alpha} \; J(\alpha) = \sum_{i=1}^{m}\alpha_i - \frac{1}{2}\sum_{i,j=1}^{m}\alpha_i\alpha_j y_i y_j \left\langle x_i, x_j \right\rangle$$

$$s.t. \quad 0 \leq \alpha_i \leq C$$

$$\sum_{i=1}^{m}\alpha_i y_i = 0$$

# WHAT IS SOFT-MARGIN SVM?

- **Multi-class SVMs**




Source:
Left image: Malhotra, V. (2020). *ML06: Intro to Multi-class Classification* [online], viewed May 24 2022, from: <https://medium.com/analytics-vidhya/ml06-intro-to-multi-class-classification-e61eb7492ffd>
Right image: Rhys, H. I. (2020). *Machine Learning with R, the tidyverse, and mlr*. USA: Manning Publications

# ADVANTAGES AND DISADVANTAGES OF SVMS

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**Advantages**

- Being a highly sophisticated and mathematically sound algorithm, SVM is one of the most accurate machine learning algorithms.
- It is a dynamic algorithm and can solve a range of problems, including linear and non-linear problems, binary, binomial, and multi-class classification problems, along with regression problems.
- SVM uses the concept of margins and tries to maximize the differentiation between two classes; it reduces the chances of model overfitting, making the model highly stable.

# ADVANTAGES AND DISADVANTAGES OF SVMS

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**Advantages**

- Because of the availability of kernels and the very fundamentals on which SVM is built, it can easily work when the data is in high dimensions and is accurate in high dimensions to the degree that it can compete with algorithms such as Naive Bayes, which specializes in dealing with classification problems of very high dimensions.
- SVM is known for its computation speed and memory management. It uses less memory, especially when compared to ***machine vs deep learning*** algorithms with whom SVM often competes and sometimes even outperforms to this day.

# ADVANTAGES AND DISADVANTAGES OF SVMS

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**Disadvantages**

- While SVM is fast and can work in high dimensions, it still loses to Naive Bayes, which provides faster predictions in high dimensions. SVM also takes a relatively long time during the training phase. Many a time before SVM modeling, you may have to use dimension reduction techniques like ***Factor Analysis or PCA (Principal Component Analysis)***.
- Like some other machine learning algorithms, which are often highly sensitive towards some of their hyper-parameters, SVM's performance is also highly dependent upon the kernel chosen by the user.
- Compared to other linear algorithms such as Linear Regression, SVM is not highly interpretable, especially when using kernels that make SVM non-linear. Thus, it is not easy to assess how the independent variables affect the target variable.
- Lastly, a good amount of computational capability might be required (especially when dealing with a huge dataset) in tuning the hyper-parameters such as the value of cost and gamma.

# APPLICATIONS OF SVM

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- **Face detection**: SVMs classify parts of the image as face and non-face and create a square boundary around the face.
- **Text and hypertext categorization**: SVMs allow text and hypertext categorization for both inductive and transductive models. They use training data to classify documents into different categories. It categorizes on the basis of the score generated and then compares with the threshold value.
- **Classification of images**: Use of SVMs provides better search accuracy for image classification. It provides better accuracy in comparison to the traditional query-based searching techniques.

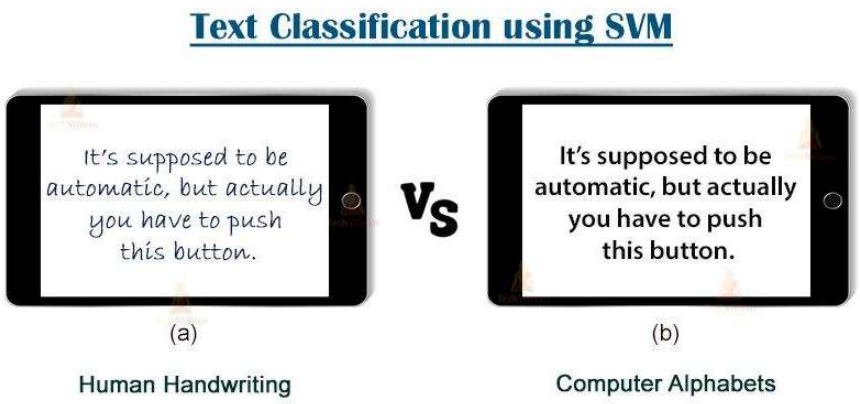
# APPLICATIONS OF SVM

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- **Bioinformatics**: Applications of SVM in bioinformatics include protein classification and cancer classification. We use SVM for identifying the classification of genes and patients on the basis of genes and other biological problems.
- **Protein fold and remote homology detection:** We can apply SVM algorithms for protein remote homology detection.
- **Handwriting recognition:** We use SVMs to recognize handwritten characters which are widely used.
- **Generalized predictive control (GPC):** We use SVM-based GPC to control chaotic dynamics with useful parameters.

# APPLICATIONS OF SVM




Source: TechVidvan. *SVM Applications – Top 10 astonishing real life applications of SVM* [online], viewed May 24 2022, from: <https://techvidvan.com/tutorials/svm-applications/>

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# Thank you!